WEEK 1 WEEKLY WORK REPORT

Lab Setup, Linux and Networking Foundations.

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1. Overview

Objective: Build a safe home cybersecurity lab and learn Linux command-line skills, networking fundamentals, and basic security concepts.

Scope: VirtualBox setup; Kali Linux, Ubuntu, and Windows evaluation VMs; Linux users/groups/permissions; networking checks; TryHackMe learning rooms; OverTheWire Bandit levels 0–10; command cheat sheet; documentation and reflection.

Status: Use this report as a working template. Replace all bracketed fields with your own verified work and mark incomplete tasks honestly.

2. Lab Environment

3. Virtualization: VMware

VMs: Kali Linux, Ubuntu, Windows 10.

Network mode: Host-only or NAT Network

Snapshots: Record the clean snapshot for each VM

Evidence: Store authentic screenshots in Findings/Lab Setup/.

3. Lab Setup and Connectivity

Configuration: Created or configured the VMs and selected an isolated virtual network. Intentionally vulnerable machines should not be bridged to the physical LAN.

Validation: Record actual output from `ip a` / `ipconfig`, `ip route`, and ping tests between the lab VMs. Windows Firewall or other controls may block ICMP, so note the observed result accurately.

Evidence checklist: VM list screenshot; each VM's IP configuration; adapter settings; successful ping screenshots where applicable; clean snapshot screenshots.

4. Linux Practice

Accounts: Three users created: [names or non-sensitive identifiers]. Two groups created: [group names].

Shared directory: Path: [path]; owner/group: [owner:group]; permissions: [mode]; access verified as: [accounts].

Commands: Document at least 30 commands with purpose and actual output. Include examples from `find`, `grep`, `ps`, `ss -tulpn`, file permissions, and network troubleshooting.

5. Network Inventory and Diagram

Inventory: For each VM, record IP/prefix, default gateway, DNS server(s), adapter mode, and date checked.

Tools: Use `ip a`, `ip route`, `ping`, `traceroute` or `tracepath`, and `nslookup` or `dig`. On Windows, use `ipconfig /all`, `ping`, and `nslookup`.

Diagram: Create a draw.io diagram showing the virtual network, each VM, actual IP addresses, and gateway/DNS where applicable. Save editable `.drawio` and export PNG/PDF.

6. Learning Platform Progress

TryHackMe: Track each room separately: Search Skills; Learning Cyber Security; Linux Fundamentals Part 1; What is Networking; Introductory Networking; Networking Concepts; Pyramid Of Pain; Intro to LAN.

OverTheWire Bandit: Track levels 0–10 individually. Record approach, commands, lessons, and evidence. Never include level passwords in writeups or screenshots.

7. Key Concepts Reviewed

Cybersecurity and CIA: Confidentiality protects information from unauthorized viewing; integrity protects accuracy and authorized changes; availability keeps services accessible when needed.

Security roles: SOC analysts monitor and investigate alerts; penetration testers assess weaknesses within an authorized scope; incident responders contain, investigate, and recover from incidents.

Networking: IP supports logical addressing and routing; MAC addresses support local-link delivery; DNS resolves names; DHCP leases configuration; ports identify service endpoints.

Linux: The filesystem hierarchy, users and groups, permissions, process inspection, logs, and scheduled jobs are core administration skills.

Risk terminology: Threat is a potential cause of harm; vulnerability is a weakness; exploit is a method that uses a weakness; risk combines likelihood and impact.

8. Security, Legal and Ethical Practice

Authorization and scope: Test only systems you own or have explicit permission to assess. Record approved targets, methods, timing, and limits.

Lab isolation: Use Host-only or a carefully configured NAT Network for vulnerable lab machines; avoid Bridged mode for intentionally vulnerable targets.

Evidence hygiene: Use genuine screenshots, minimize collected data, redact unrelated personal information, and do not publish passwords, keys, tokens, or Bandit credentials.

Legal note: For an India-based lab, the Information Technology Act, 2000, as amended, may be relevant to unauthorized access and computer-related offenses. This is educational information, not legal advice; verify current official law.

9. Conclusion and Next Steps

Conclusion: [Summarize what you can now explain or do independently, supported by evidence.]

Next steps: Complete remaining learning rooms and Bandit levels; validate VM connectivity and isolation; finish the command log; export the network diagram; review report evidence; and confirm submission requirements with the coordinator.

Appendix - Evidence index

Evidence category	Expected files / details
Lab setup	VM list, adapter settings, IP details, ping tests, clean snapshots
Linux practice	User/group setup, shared folder permissions, 30 commands with actual output, cron evidence
Network map	IP/gateway/DNS inventory, draw.io source file, PNG/PDF export
TryHackMe	Room writeups and completion screenshots where available
Bandit	Level 0–10 progress evidence; no passwords
Reports	Final PDF, presentation, README, cheat sheet